Generating Product Images from Customer Reviews

Artifact-first, review-grounded visual profiles for dual-provider image generation and evaluation
Repository (GitHub): https://github.com/xinxuxin/ai-lab-final
Repository snapshot (commit): `20e58392dfe1916ff54cd814932c99555b81394b`
Local artifact root (this submission): `ai-lab-final/`

1. Final Report Requirements & Coverage (per assignment)

The assignment requires the final submission to include:

1. **A final report documenting analyses, findings, and explanations**
   - Covered in: Sections **3–8** (Data, Method, Workflow, Evaluation, Results)
2. **Links/files for collected data, written code, designed prompts, generated images**
   - Covered in: Section **10 Reproducibility & Artifact Index** + Appendix links
   - All artifacts are included under `ai-lab-final/` (data, outputs, prompts, docs).
3. **Reflective discussion on challenges faced and lessons learned**
   - Covered in: Section **9 Reflective Discussion**

2. Project Overview

Research question:

> Can grounded textual evidence from product listings and customer reviews improve the quality of generated product imagery compared with using product descriptions alone?

Key design principles

- **Artifact-first**: each stage writes durable artifacts (data, profiles, prompts, images, evaluation outputs) that can be inspected and reused.
- **Inspectability**: use a staged prompt chain and persist retrieval evidence and traces to local files.
- **Reproducibility**: strict verification via `verify-artifacts --stage full` and a submission package builder.

3. Data & Product Selection (Q1)

We select **exactly three products** from **distinct categories**, scrape their public description/reviews/images once, and validate cleaned artifacts.
Selected products and Q1 validation summary (from `ai-lab-final/docs/q1_summary.md`):

Product	Category	Product slug	Cleaned reviews	Valid real images	Description chars	Q1 checks
JBL Tour One M2 Wireless Over-Ear ANC Headphones (Black)	audio	`jbl-tour-one-m2-wireless-over-ear-adaptive-noise-cancelling-headphones-black-89301002`	8	10	3453	PASS
Levoit Core 300 Air Purifier White	home-appliance	`levoit-core-300-air-purifier-white-81910071`	8	14	475	PASS
Desk Lamp with USB Ports (Threshold™)	lighting	`desk-lamp-with-usb-ports-includes-led-light-bulb-threshold-8482-80705997`	7	4	604	PASS

Artifact locations

- Raw scrape artifacts: `ai-lab-final/data/raw/<product_slug>/`
- Cleaned corpora: `ai-lab-final/data/processed/<product_slug>/`
- Q1 validation: `ai-lab-final/backend` CLI `verify-artifacts --stage q1` (see Section 10)

4. Method (Q2 + Q3)

4.1 Two modes: baseline vs. review-informed

We implement two analysis modes to isolate the impact of reviews:

1. `baseline_description_only`
    - Uses only the cleaned product description as the source of truth.
2. `review_informed_rag`
    - Uses cleaned reviews and retrieval evidence to surface grounded details, especially **expectation-vs-reality** cues.

Outputs are written as structured `VisualProfile` JSON artifacts under:

- `ai-lab-final/outputs/visual_profiles/<product_slug>/baseline_description_only.json`
- `ai-lab-final/outputs/visual_profiles/<product_slug>/review_informed_rag.json`
- `ai-lab-final/outputs/visual_profiles/<product_slug>/retrieval_evidence.json`

4.2 Chunking strategy

We treat **one review as the default chunk unit** (splitting only overly long reviews). This preserves interpretability: each retrieved snippet can be traced back to a concrete review.

4.3 Retrieval design (review-informed mode)

Retrieval is embedding-based by default (OpenAI embeddings), with fallback keyword overlap retrieval if embeddings are unavailable. We issue **aspect-specific queries**, such as:

- appearance / shape
- color / finish
- material / texture
- size / scale
- expectation vs. reality mismatches

4.4 Prompt chain and VisualProfile schema

Instead of a monolithic prompt, we run a staged chain:

1. **Aspect evidence extraction**
2. **Conflict resolution**
3. **Final VisualProfile synthesis**

The VisualProfile schema includes:

- `high_confidence_visual_attributes`
- `low_confidence_or_conflicting_attributes`
- `common_mismatches_between_expectation_and_reality`
- `prompt_ready_description`
- `negative_constraints`

Prompt templates (repo):

- Q2 prompts: `ai-lab-final/prompts/q2/`
- Q3 prompts: `ai-lab-final/prompts/q3/`

5. Image Generation (Q3)

5.1 Providers

We use two API-only image-generation providers:

- **OpenAI Images API**
- **Stability AI API**

5.2 Prompt flow (pilot → final)

For each product and each provider, we generate:

1. 1 pilot prompt + 1 pilot image
2. refine into a final prompt
3. 4 final images

This yields:

- 5 images / provider / product
- 3 products × 2 providers × 5 images = **30 generated images**

Outputs are stored under:

- `ai-lab-final/outputs/generated_images/<product_slug>/<provider>/pilot/`
- `ai-lab-final/outputs/generated_images/<product_slug>/<provider>/final/`
- `ai-lab-final/outputs/generated_images/<product_slug>/<provider>/prompt_versions.json`

5.3 Prompt constraints

Prompt construction enforces:

- single product only, centered composition
- neutral/studio background
- realistic product photography
- no unsupported accessories
- negative constraints (when supported)

6. Agentic Workflow (Q4)

We build a typed multi-agent workflow that connects Q1 → Q4 stages with explicit responsibilities and durable handoffs.

6.1 Workflow

Instead of a single monolithic prompt, we decompose the project into six typed agents. Each agent consumes explicit upstream artifacts, decides whether to reuse or recompute outputs, and writes durable handoff artifacts for downstream stages:

1. **DataCurationAgent (Q1)**
    - Inputs: raw scrape artifacts (`data/raw/`)
    - Outputs: processed corpora (`data/processed/`) + Q1 validation pass/fail state

2. **RetrievalAgent (Q2 evidence layer)**
    - Inputs: processed reviews (review-level chunks)
    - Outputs: aspect-specific retrieval evidence + LLM trace artifacts
3. **VisualUnderstandingAgent (Q2 synthesis)**
    - Inputs: description-only text (baseline) + retrieved review evidence (RAG mode)
    - Outputs: two VisualProfiles per product (`baseline_description_only`, `review_informed_rag`)
4. **PromptComposerAgent (Q3 prompt planning)**
    - Inputs: VisualProfiles + prompt templates
    - Outputs: pilot/final prompt versions (with lineage)
5. **ImageGenerationAgent (Q3 generation)**
    - Inputs: prompt versions
    - Outputs: pilot image + final image set for each provider (OpenAI / Stability) + generation manifests
6. **EvaluationAgent (Q3 comparison + analytics)**
    - Inputs: real reference images + generated images
    - Outputs: comparison/evaluation artifacts (human scoring sheet, summaries, optional vision-assisted results)

6.2 Saved workflow trace artifacts

Example saved run (from `ai-lab-final/docs/master_guide.md`):

- run id: `20260420T160007Z-a7fac78d`
- `ai-lab-final/outputs/workflow_runs/20260420T160007Z-a7fac78d/trace.json`
- `ai-lab-final/outputs/workflow_runs/20260420T160007Z-a7fac78d/stage_status.json`
- `ai-lab-final/outputs/workflow_runs/20260420T160007Z-a7fac78d/artifact_links.json`

7. Evaluation Protocol

7.1 Rubric (6 dimensions)

We evaluate real-vs-generated fidelity using six dimensions:

1. color accuracy
2. material / finish accuracy
3. shape / silhouette accuracy
4. component completeness
5. size / proportion impression
6. overall recognizability

7.2 Evaluation modes

1. **Human scoring template** (`human_score_sheet.csv`)
2. **Optional vision-assisted evaluation** (`vision_assisted_eval.json`)

Note: some products may show **partial** completion in the vision-assisted layer due to rate limits; partial outputs are preserved and used transparently (see Section 9 and repo docs).
Evaluation artifacts per product:

- `ai-lab-final/outputs/evaluations/<product_slug>/summary.json`
- `ai-lab-final/outputs/evaluations/<product_slug>/human_score_sheet.csv`
- `ai-lab-final/outputs/evaluations/<product_slug>/vision_assisted_eval.json`

8. Results & Findings

8.1 Qualitative comparison panels (real vs. generated)

Below are representative comparisons used in our analysis:

JBL Headphones (strongest overall)

Real

OpenAI

Stability

Levoit Air Purifier (minimal geometry; mid difficulty)

Real

OpenAI

Stability

Desk Lamp (hardest; multi-part structure)

Real

OpenAI

Stability

8.2 Quantitative summaries (from `docs/master_guide.md`)

Product 1: JBL Headphones (vision-assisted: completed)

Dimension	OpenAI	Stability
color accuracy	4.50	4.25
material finish accuracy	4.00	3.75
shape silhouette accuracy	3.50	4.00
component completeness	3.00	3.25
size proportion impression	4.00	4.00
overall recognizability	3.75	4.00

Product 2: Levoit Air Purifier (vision-assisted: partial)

Dimension	OpenAI	Stability
color accuracy	4.00	4.00
material finish accuracy	3.75	3.00
shape silhouette accuracy	3.75	2.00
component completeness	3.25	3.00
size proportion impression	4.00	2.00
overall recognizability	3.50	3.00

Product 3: Desk Lamp (vision-assisted: completed)

Dimension	OpenAI	Stability
color accuracy	2.75	2.00
material finish accuracy	2.50	1.75
shape silhouette accuracy	2.25	1.50
component completeness	3.00	1.00
size proportion impression	2.75	1.75
overall recognizability	2.50	1.50

8.3 Cross-product insights

From the saved artifact snapshot:

- **Category drives difficulty** more strongly than provider choice or review richness.
  - Headphones were easiest.
  - Desk lamp (multi-part geometry) was hardest.
  - Minimal appliances sat in the middle.
- **OpenAI is stronger overall** across the saved evaluation outputs.
- **Failure modes differ**:
  - OpenAI: more issues with *style consistency.*
  - Stability: more issues with *structural accuracy.*

9. Reflective Discussion (Challenges & Lessons Learned)

9.1 Challenges

1. **Partial vision-assisted evaluation for minimal products**
   For the Levoit purifier, minimal surface detail made some rubric dimensions harder to score consistently; partial vision-model assistance was used to resolve disputed scores.
2. **Rate-limit sensitivity**
   Both generation APIs enforce tight per-minute request/token caps, requiring careful retry logic and pacing across the full 30-image run.
3. **Review availability constraints**
   Review collection depends on what the public product payload exposes; some products had fewer accessible reviews than expected.
4. **Structural products remain hard to render**
   Products with precise geometry and multiple parts (desk lamp) exposed limitations of diffusion-style generation when part count and angles matter.

9.2 Lessons learned

1. **Reviews add visible detail**
   Customer text often surfaces concrete cues (e.g., color discrepancies, surface texture, scale) that official descriptions omit.
2. **Category is the strongest predictor of generation quality**
   Product type mattered more than provider choice or review richness in this snapshot.
3. **Models fail differently**
   Understanding failure modes can be as important as comparing average scores.
4. **Evaluation should preserve cues, not just "visual appeal"**
   A defensible evaluation asks whether generated images preserve **high-confidence visual attributes** extracted from the profile.

9.3 In-class Q&A reflection: Why six evaluation dimensions (and why not CLIP score)?

During our in-class presentation, a classmate asked:

> **How were the six evaluation dimensions designed? Why not use automated metrics like CLIP score?**

We give a more complete answer here. We designed the six evaluation dimensions to be **attribute-level**, **diagnostic**, and directly aligned with the goal of this project: checking whether generated images preserve the **high-confidence cues** extracted in our `VisualProfile` (rather than only matching the "general idea" of the product). Concretely, the dimensions were chosen to cover complementary failure modes:

1. **Color accuracy** (e.g., black vs. white variants, accent colors)
2. **Material / finish accuracy** (e.g., matte plastic vs. metallic sheen, fabric vs. leather)
3. **Shape / silhouette accuracy** (global geometry; does the outline match the real product?)
4. **Component completeness** (are critical parts present and correctly counted/placed—ports, buttons, lamp arm/base, ear-cups, etc.?)
5. **Size / proportion impression** (relative scale and proportions between parts)
6. **Overall recognizability** (would a viewer identify this as the intended product at a glance?)

We intentionally did **not** use automated metrics such as **CLIP score** as our primary measure because they mainly capture **holistic semantic similarity** between an image and text (or between two images). In product imagery, that is often insufficient: an image can score well on "headphones" while still being wrong in **part count**, **component placement**, or **variant-specific color/finish**, which are precisely the mistakes we want to surface. Instead, we use **human scoring** along these six dimensions (with **partial vision-model assistance** only for ambiguous cases) to produce evidence that is both **interpretable** (you can see which dimension failed) and **actionable** (it suggests whether to fix retrieval/profile extraction vs. prompt constraints vs. provider choice).

9.4 Final takeaway

- **Review-informed pipelines are more rigorous**: grounding in review evidence yields more accurate and defensible outputs than descriptions alone.
- **Multi-stage analysis improves inspectability**: separating extraction, conflict resolution, and synthesis makes decisions traceable and correctable.
- **Reproducible and presentation-ready**: artifact-first design, strict verification, and a full frontend demo

make the work verifiable end-to-end.

10. Reproducibility & Artifact Index

10.1 Environment variables (required for full reruns)

Copy `ai-lab-final/.env.example` to `ai-lab-final/.env` and set:

```
OPENAI_API_KEY=...
STABILITY_API_KEY=...
```

10.2 Minimal command set (relative paths)

From `ai-lab-final/`:

```
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -e "backend[dev]"
cp .env.example .env
```

Verify saved artifacts (no regeneration required):

```
cd backend
source ../.venv/bin/activate
PYTHONPATH=src ../.venv/bin/python -m cli.main verify-artifacts --stage full --frontend-dir ../frontend
```

10.3 Artifact index table

Stage	What it contains	Key paths (local)
Q1	discovery + raw scrape + processed corpora	`ai-lab-final/data/discovery/`, `ai-lab-final/data/raw/`, `ai-lab-final/data/processed/`
Q2	baseline + review-informed VisualProfiles + retrieval evidence	`ai-lab-final/outputs/visual_profiles/`, `ai-lab-final/prompts/q2/`
Q3	pilot/final prompts + generated images (OpenAI/Stability)	`ai-lab-final/outputs/generated_images/`, `ai-lab-final/prompts/q3/`
Q3 eval	human scoring sheet + vision-assisted summaries	`ai-lab-final/outputs/evaluations/`
Q4	agentic workflow traces + artifact links	`ai-lab-final/outputs/workflow_runs/`, `ai-lab-final/docs/agentic_workflow.md`

Appendix A — Prompt templates (links in this package)

Q2 prompt chain:

- `ai-lab-final/prompts/q2/aspect_evidence_extraction.md`
- `ai-lab-final/prompts/q2/conflict_resolution.md`
- `ai-lab-final/prompts/q2/final_visual_profile_synthesis.md`

Q3 prompts:

- `ai-lab-final/prompts/q3/pilot_prompt.md`
- `ai-lab-final/prompts/q3/final_prompt.md`